

# Documentation Week 1 - Liv

## Presentation Link

📄 STAT 390: Presentation 1

## Dashboard/Code Link

[Dashboard](#)

[CAR Dashboard](#)

## What I Did

Our team is developing a Power BI dashboard to analyze Legal Aid's phone system and identify inefficiencies in how clients navigate automated menus. The goal is to reduce caller frustration and support data-driven decisions for simplifying call flows.

My key contributions included:

- New Metric Ideation: Helped design and define Submenu Efficiency (average number of steps before failure).
  - Currently, this variable measures the number of menus navigated per call.
- Power BI Support: Helped group members import data and start exploratory visuals.
- Data Cleaning/EDA (All Calls Dataset):
  - Edwin and I worked on the following:
  - Converted the Called number column to Text, since it contained special characters (#) that caused Power BI import errors.
  - Removed all records with Duration = 0 to prevent skewed averages.
  - Standardized Contact Session ID as a consistent session identifier, since External Caller ID varied between calls.
- Exploratory Analysis (CAR Dataset):
  - I worked on the following on my own:
  - Created a variable to count how many menus are navigated.
    - Using COUNTIF formula in Excel
  - Clustered Column Charts:
    - Count of Calls by Average Number of Submenus Navigated
    - Average Number of Submenus Navigated by Activity Type

## How This Helped the Project

- Columns Used:
  - From All Calls Dataset: Duration, Redirect reason, Call outcome, User type, Answer indicator.

- From CAR Dataset: Contact Session ID, Activity Name, Menus Navigated (new).
- Insights Derived:
  - “Deflection” and “Unconditional” redirect reasons have the longest average durations, suggesting possible routing inefficiencies.
  - “FollowMe” and “NoAnswer” are most frequent redirect reasons but have shorter durations, indicating quick hangups or missed calls.
  - Automated systems (like VirtualLine and AutomatedAttendantVideo) tend to have shorter call times, likely due to quick routing or early disconnections.
  - These findings helped guide metric design (Submenu Efficiency and Redirection Performance) which quantify caller effort and system success.
  - A better understanding of the average amount of submenus navigated.
  - Which activities are creating the most complication in having to navigate many submenus.
- Questions for Cynthia:
  - In the Redirect Reason column, what does an “N/A” value represent?
    - Does it mean the call was never redirected, or that the redirect data wasn’t captured for that session?
  - When a call ends in a ClosedQueueMenu, does Webex record that as a “failed” or “closed” outcome in the data, or is that outcome only visible in the CAR dataset?
  - For Duration, does that value reflect the entire call length (including ringing and menu navigation), or only the time once the call is answered or queued?

## Issues Faced & Resolutions

| Issue               | Cause                                                            | Fix                                                 |
|---------------------|------------------------------------------------------------------|-----------------------------------------------------|
| Data Type Errors    | Called number contained symbols like #, breaking Power BI import | Converted to Text in Power Query                    |
| Zero-Duration Calls | Invalid data rows                                                | Filtered out during import                          |
| Unstable Caller IDs | External Caller ID changed per call                              | Used Contact Session ID for unique session tracking |

|                          |                                                                   |                                                                     |
|--------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|
| Power BI Web Limitations | Couldn't change column data types or apply transformations online | Switched to Power BI Desktop (on Edwin's PC) for full functionality |
|--------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------|

## Next Steps

- Expand EDA using both All Calls and CAR datasets to analyze:
  - Call outcomes, duration distributions, and submenu usage.
- Refine metrics:
  - Submenu Efficiency – average steps before a failed call.
    - Create a graph showing average submenus navigated to and if there are any connections between the amount of submenus navigated to with other variables.
    - Make the variable more complex by adding filters for EP Name, Flow Name, and Activity Name.
  - Redirection Performance – % of redirected calls successfully connected.
- Add new metrics:
  - Closed Queue Exposure Rate – % of calls ending in Closed Queue.
  - Navigation Time Wasted – avg. time spent before reaching a dead end.
- Build final dashboard model with improved filters and visual interactivity.

## Summary

This phase focused on cleaning data, exploring redirection trends, and defining new performance metrics to measure system efficiency. This was mostly a more basic dive into the data and figuring out Power BI, but the findings provide a foundation for deeper analysis in Presentation 2, by connecting caller navigation patterns to specific bottlenecks in Legal Aid's phone system.